

ECN

ENTERPRISE CAMPUS NETWORK

Secure · Redundant · Segmented

TECHNICAL DOCUMENTATION

HSRP REDUNDANT CAMPUS NETWORK

Cisco Packet Tracer project

ARCHITECTURE	Core / Access
REDUNDANCY	HSRP, PVST and EtherChannel
SEGMENTATION	VLAN 10, 20 and 30
SECURITY	SSH, AAA, PortFast and BPDU Guard

Document control

Field	Value
Document title	Technical Documentation - HSRP Redundant Campus Network
Version	1.0
Date	6 August 2026
Status	Final technical analysis of the supplied project
Platform	Cisco Packet Tracer
Covered devices	CORESW1, CORE2, ACCESS1 and ACCESS2
Sources	Lab05_HSRP.pkt and running-config/operational outputs from four switches

SCOPE AND RELIABILITY

Addressing, HSRP roles, VLANs, trunk links, EtherChannel and STP state were derived directly from the supplied operational outputs. Exact switch models and end-device addresses are not present in the text exports and are therefore not presented as confirmed facts. The topology was reconstructed from interface descriptions and Port-channel relationships.

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1. Introduction and project description

Lab05_HSRP is a simulated campus switching project whose primary goal is to provide a redundant default gateway for three business VLAN segments. The network contains two multilayer core switches and two Layer 2 access switches. CORESW1 and CORE2 perform inter-VLAN routing through SVI interfaces, while end devices connect through ACCESS1 and ACCESS2.

Each VLAN has an HSRP virtual address ending in .1. CORESW1 uses physical SVI addresses ending in .2, HSRP priority 110 and preemption, making it the active gateway. CORE2 uses addresses ending in .3, the default priority of 100 and preemption, making it the standby gateway. Clients therefore use a stable gateway that is independent of either physical SVI address.

The core switches are connected by an LACP EtherChannel built from two GigabitEthernet links. Each access switch uses a dual-FastEthernet EtherChannel uplink to its attached core switch. VLAN traffic is carried over IEEE 802.1Q trunks, while user-facing ports are assigned to the Management, Sales and IT VLANs.

SOLUTION SUMMARY

The project successfully demonstrates an HSRP active/standby gateway, inter-VLAN routing, LACP link aggregation, VLAN segmentation, PVST and basic access-port hardening.

2. Project objectives

- Separate Management, Sales and IT users into independent broadcast domains.
- Provide inter-VLAN routing through multilayer core switches.
- Provide one stable virtual gateway address by using HSRP.
- Automatically return CORESW1 to the active HSRP role through preemption.
- Increase uplink capacity and physical-link resilience through LACP EtherChannel.
- Prevent switching loops with PVST and accelerate edge-port activation with PortFast.
- Protect access ports from unexpected BPDU frames through BPDU Guard.
- Provide remote management of the core devices through SSHv2 and a local AAA account.

3. Network architecture and topology

Logical HSRP network topology

Two core switches, three VLANs and LACP EtherChannel uplinks

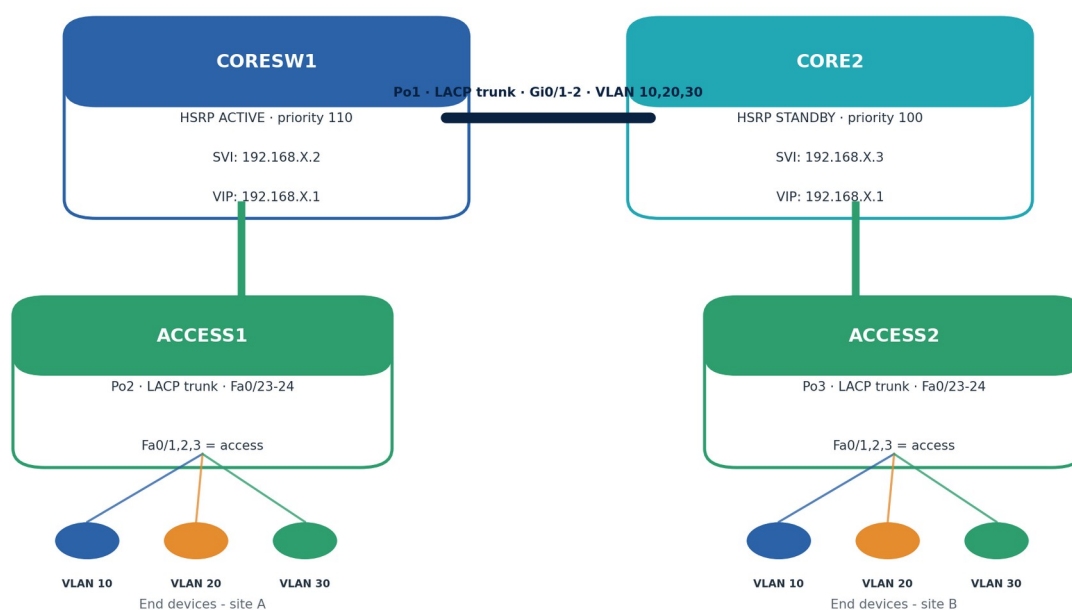


Figure 1 - Reconstructed logical topology of the HSRP network

The architecture contains a core/distribution layer and an access layer. Po1 is the shared Layer 2 trunk between the core devices. Po2 connects CORESW1 to ACCESS1, while Po3 connects CORE2 to ACCESS2. All three Port-channel interfaces carry only VLANs 10, 20 and 30; the native VLAN remains VLAN 1, but its SVI is administratively disabled.

IMPORTANT REDUNDANCY NOTE

EtherChannel protects against failure of one physical member, but each access switch is connected to only one core device. Complete failure of CORESW1 isolates ACCESS1, while complete failure of CORE2 isolates ACCESS2. HSRP therefore provides gateway redundancy only while an Layer 2 path to both core SVIs still exists.

4. Devices used

Identifier	Type	Role	IOS	Key functions
CORESW1	Multilayer switch*	Primary core/distribution	12.2(37)SE1	SVIs, HSRP active, inter-VLAN routing, PVST, LACP, SSH/AAA
CORE2	Multilayer switch*	Secondary core/distribution	12.2(37)SE1	SVIs, HSRP standby, inter-VLAN routing, PVST, LACP, SSH/AAA
ACCESS1	Layer 2 access switch*	Site A	15.0	Access ports, Po2 LACP trunk, PortFast, BPDU Guard
ACCESS2	Layer 2 access switch*	Site B	15.0	Access ports, Po3 LACP trunk, PortFast, BPDU Guard

Identifier	Type	Role	IOS	Key functions
End devices	Packet Tracer PCs	VLAN 10/20/30 users	Not applicable	Ethernet access; exact IP addresses are not present in text exports

* The exact switch model is not explicitly stated. The interface count and IOS versions match typical Packet Tracer multilayer and access devices, but the model is not presented as a confirmed fact.

5. Technologies used

Technology/protocol	Role in the project
VLAN 10, 20 and 30	Logical segmentation of Management, Sales and IT users.
IEEE 802.1Q trunking	Transport of multiple VLANs across Po1, Po2 and Po3.
SVIs and inter-VLAN routing	Layer 3 gateway interfaces on both core switches.
HSRP	Virtual gateway .1; CORESW1 active, CORE2 standby.
LACP EtherChannel	Po1 between the core devices; Po2 and Po3 as access uplinks.
PVST	A separate spanning-tree instance for each VLAN.
PortFast	Rapid transition of end-device access ports to forwarding.
BPDUGuard	Automatic edge-port protection against an attached switch.
SSHv2 and local AAA	Remote management without Telnet.
Administrative port shutdown	Fa0/4-22 on access devices and most unused core ports are shut down.

6. Addressing table

Segment	Network/prefix	Mask	Virtual gateway	CORESW1	CORE2	Note
VLAN 10 - Management	192.168.10.0/24	255.255.255.0	192.168.10.1	192.168.10.2	192.168.10.3	HSRP group 10
VLAN 20 - Sales	192.168.20.0/24	255.255.255.0	192.168.20.1	192.168.20.2	192.168.20.3	HSRP group 20
VLAN 30 - IT	192.168.30.0/24	255.255.255.0	192.168.30.1	192.168.30.2	192.168.30.3	HSRP group 30
ACCESS1 management	Not defined	-	-	-	-	No active management SVI
ACCESS2 management	Not defined	-	-	-	-	No active management SVI

END-DEVICE ADDRESS NOTE

The supplied text exports do not show end-device IP addresses, DHCP pools or default-gateway settings. The table therefore includes only confirmed network, SVI and HSRP addresses.

7. VLAN and port plan

VLAN	Name	Subnet	Gateway	ACCESS1	ACCESS2	Purpose
10	Management	192.168.10.0/24	192.168.10.1	Fa0/1	Fa0/1	Management users/devices
20	Sales	192.168.20.0/24	192.168.20.1	Fa0/2	Fa0/2	Sales department
30	IT	192.168.30.0/24	192.168.30.1	Fa0/3	Fa0/3	IT department

7.1. Uplink plan

Link	Logical interface	Physical members	Mode	Allowed VLANs
CORESW1 ↔ CORE2	Port-channel1	Gi0/1, Gi0/2 na oba uređaja	LACP active; 802.1Q trunk	10,20,30
CORESW1 ↔ ACCESS1	Port-channel2	Fa0/23, Fa0/24 na oba uređaja	LACP active; 802.1Q trunk	10,20,30
CORE2 ↔ ACCESS2	Port-channel3	Fa0/23, Fa0/24 na oba uređaja	LACP active; 802.1Q trunk	10,20,30

8. Detailed network diagram

Detailed L2/L3 diagram and addressing plan

HSRP virtual address .1 is the default gateway in every VLAN

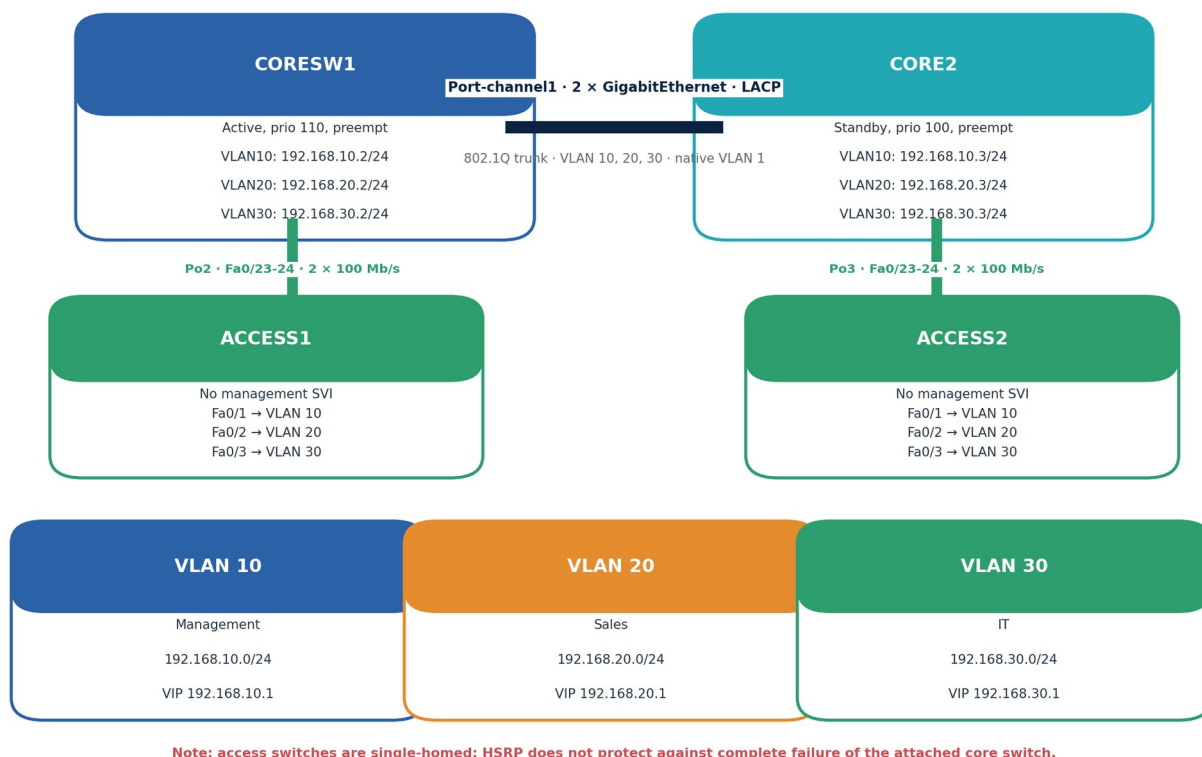


Figure 2 - Detailed L2/L3 diagram with ports, addresses and VLANs

9. Device configuration analysis

9.1. CORESW1 - primary gateway

CORESW1 has SVI addresses 192.168.10.2, 192.168.20.2 and 192.168.30.2. The operational HSRP output confirms that the local device is Active in groups 10, 20 and 30, with priority 110 and preemption enabled. The standby peer is CORE2 at the .3 addresses.

- Po1 is an active LACP trunk to CORE2 over Gi0/1 and Gi0/2.
- Po2 is an active LACP trunk to ACCESS1 over Fa0/23 and Fa0/24.
- The trunk allowed list is limited to VLANs 10, 20 and 30.
- Most unused FastEthernet ports are administratively shut down.
- SSHv2 and a local AAA user are configured; VTY lines accept SSH only.

```
show standby brief
Vl10  10  110 P Active local 192.168.10.3 192.168.10.1
Vl20  20  110 P Active local 192.168.20.3 192.168.20.1
Vl30  30  110 P Active local 192.168.30.3 192.168.30.1
```

9.2. CORE2 - secondary gateway

CORE2 has SVI addresses ending in .3. The HSRP output confirms the Standby state and identifies CORESW1 as the active peer. The default HSRP priority of 100 is lower than the value 110 on CORESW1, so CORE2 does not take the active role during normal operation.

- Po1 connects CORE2 to CORESW1 through two GigabitEthernet members.
- Po3 connects CORE2 to ACCESS2 through Fa0/23 and Fa0/24.
- Directly connected routes exist for all three /24 VLAN subnets.
- SSHv2, local AAA and shutdown of unused ports are applied as on CORESW1.

9.3. ACCESS1 and ACCESS2

Both access devices use the same user-port layout: Fa0/1 belongs to VLAN 10, Fa0/2 to VLAN 20 and Fa0/3 to VLAN 30. PortFast and BPDU Guard are enabled on all three ports. Fa0/4-22 are shut down. Uplinks Fa0/23-24 are bundled into Po2 or Po3.

ACCESS-SWITCH MANAGEMENT

Although SSH and AAA are configured, ACCESS1 and ACCESS2 have no active IP-addressed SVI and no ip default-gateway. They are therefore not remotely reachable over IP. For management, for example, add VLAN 10 SVI addresses 192.168.10.4/24 and 192.168.10.5/24 and default gateway 192.168.10.1.

10. Problem and solution

10.1. Observed problem: HSRP/STP misalignment

The HSRP operational state is clear: CORESW1 is Active for all three VLANs. However, the show spanning-tree outputs do not reflect the configured priority commands. VLAN 10 and VLAN 20 identify CORE2 as the root bridge, while VLAN 30 identifies ACCESS1 as the root. At the same time, the CORESW1 running configuration contains priority 24576 and CORE2 contains priority 28672 for VLANs 1,10,20,30. This difference between the intended configuration and the operational state indicates an unapplied/unrefreshed STP configuration or a Packet Tracer simulation inconsistency.

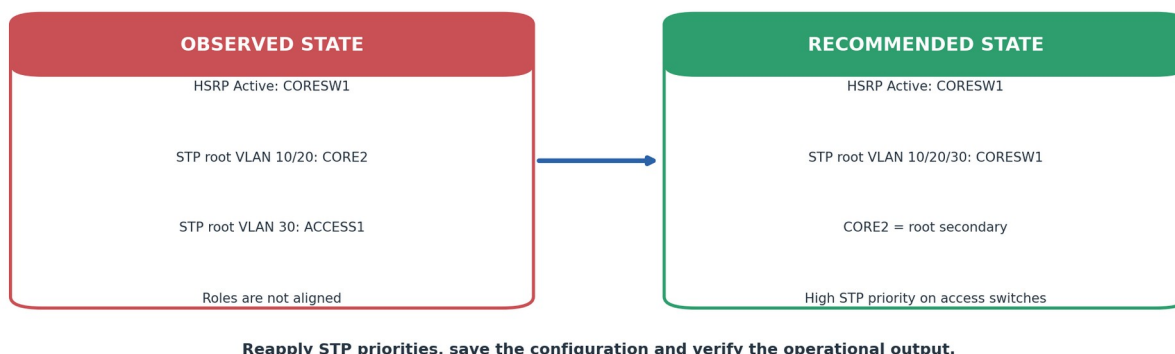


Figure 3 - Comparison of observed and recommended STP/HSRP state

- The HSRP active gateway and STP root are not located on the same core device.
- An access switch becomes the root for VLAN 30, which is undesirable in a hierarchical campus design.
- Convergence and traffic paths become less predictable.
- The operational state cannot be reliably inferred from running-config alone.

10.2. Recommended solution

The primary goal is to make CORESW1 both the HSRP Active gateway and the STP root primary for VLANs 10, 20 and 30, with CORE2 as root secondary. Access devices should retain the default or an explicitly higher STP priority. The commands should be reapplied, the configuration saved and the operational output verified.

```

CORESW1(config)# spanning-tree vlan 10,20,30 root primary
CORE2(config)# spanning-tree vlan 10,20,30 root secondary
ACCESS1(config)# spanning-tree vlan 10,20,30 priority 61440
ACCESS2(config)# spanning-tree vlan 10,20,30 priority 61440
!
copy running-config startup-config
show spanning-tree vlan 10
show spanning-tree vlan 20
show spanning-tree vlan 30
show standby brief

```

If Packet Tracer continues to display stale state, remove and reapply the priority commands, briefly shut/no shut the Port-channel interfaces or reload the project. In a real IOS environment, the operational show output remains the final source of truth.

10.3. Architectural improvement

For complete protection against a core-device failure, ACCESS1 and ACCESS2 should be connected to both core switches. In Packet Tracer this can be implemented with separate trunks controlled by STP, while a production design may use a switch stack, VSS/StackWise Virtual or MLAG/vPC, depending on the platform.

11. Traffic flows, security and redundancy

11.1. Same-VLAN communication

When two users belong to the same VLAN, traffic remains at Layer 2. If they are connected to different access switches, the frame traverses the access uplink, the inter-core Po1 trunk and the second access uplink.

11.2. Inter-VLAN communication

A client sends a packet to the HSRP virtual address 192.168.X.1. During normal operation, CORESW1 receives the packet through the HSRP virtual MAC address, routes it to the destination SVI and forwards it into the required VLAN. If CORESW1 loses the HSRP function, CORE2 takes over the virtual address without any gateway change on the client.

11.3. Security controls

- SSHv2 is the only permitted protocol on VTY lines 0-4.
- AAA uses the local account database; a privileged user exists on the core devices.
- PortFast is applied only to end-device access ports.
- BPDU Guard disables an access port that receives a BPDU, reducing the risk of an unauthorized switch connection.
- Unused ports are administratively shut down.
- DTP negotiation is disabled with switchport nonegotiate on trunk links.

ADDITIONAL RECOMMENDATION

For a production network, add dedicated management SVI addresses to the access devices, restrict SSH access with a standard ACL, use a dedicated native/parking VLAN instead of VLAN 1 and enable port security where appropriate.

12. Test plan and recommendations

Area	Command/procedure	Expected result
VLAN and trunk	show vlan brief; show interfaces trunk	VLANs 10/20/30 exist; Po1/Po2/Po3 are trunking and allow only 10,20,30.
EtherChannel	show etherchannel summary	Po1, Po2 and Po3 show SU; both members are marked P.
HSRP	show standby brief	CORESW1 Active prio 110; CORE2 Standby prio 100; VIP .1.
STP	show spanning-tree vlan 10,20,30	CORESW1 root primary; CORE2 root secondary; access devices are not root.
Inter-VLAN	ping between VLANs 10, 20 and 30	Packets are routed through the HSRP gateway.
Member failure	shut down one EtherChannel member	The Port-channel remains up through the second member.
HSRP failover	shut down an SVI on CORESW1	CORE2 becomes Active; the client gateway remains .1.
SSH	SSH to the core SVI addresses	Access works with the local AAA account; Telnet is rejected.

12.1. Priority recommendations

Priority	Finding	Recommendation
High	Operational STP roles do not match the intended configuration.	Reapply root primary/secondary, raise access-switch priority and verify show outputs.
High	Access switches are single-homed.	Connect each access device to both core switches.
Medium	ACCESS1/2 have no management SVI.	Assign VLAN 10 addresses and ip default-gateway 192.168.10.1.
Medium	The native VLAN is VLAN 1.	Use a dedicated unused native VLAN and match both ends of each trunk.
Low	CORESW1 and CORE2 naming is not standardized.	Standardize naming convention, interface descriptions and documentation.

13. Conclusion

Lab05_HSRP is a functional educational project that combines the key concepts of a redundant campus switching design: VLAN segmentation, inter-VLAN routing, HSRP, LACP EtherChannel, PVST, PortFast, BPDU Guard and more secure SSH management.

The HSRP function is operationally confirmed: CORESW1 is Active with priority 110, CORE2 is Standby with priority 100, and virtual addresses 192.168.10.1, 192.168.20.1 and 192.168.30.1 provide stable user gateways. EtherChannel links are active and both physical members participate in each bundle.

The most important finding is the mismatch between the configured STP priorities and the displayed operational root state. Before production use, the HSRP and STP roles should be aligned, access devices should be prevented from becoming root, management addressing should be added to the access switches and dual-homing to both core devices should be implemented. After those corrections, the project provides a significantly more reliable foundation for a redundant business LAN.

FINAL ASSESSMENT

The project demonstrates HSRP and Layer 2 link redundancy well. Full device redundancy and a predictable control plane require the STP corrections and a dual-homed access design.

Appendix A - Sources and key commands

Source	Content
Lab05_HSRP.pkt	Cisco Packet Tracer project file.
SWCORE 1(1).txt	CORESW1 running-config, STP, EtherChannel, trunk, HSRP, routes and SVI state.
SWCORE2(1).txt	CORE2 running-config, STP, EtherChannel, trunk, HSRP, routes and SVI state.
ACCESS 1(2).txt	ACCESS1 running-config, STP, trunk and EtherChannel state.
ACCESS 2(2).txt	ACCESS2 running-config, STP, trunk and EtherChannel state.

Use the following commands for final project verification:

```
show ip interface brief
show standby brief
show vlan brief
show interfaces trunk
show etherchannel summary
show spanning-tree vlan 10
show spanning-tree vlan 20
show spanning-tree vlan 30
show ip route
```